Communication–Localization Fingerprint-Aided Multi-UAV Trajectory Planning in Urban Environments

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Abstract—Building blockage in urban low-altitude environments causes spatial discontinuity in communication and localization service quality between uncrewed aerial vehicles (UAVs) and ground base stations. Even with integrated localization and communication (ILAC), communication outages and localization lock-loss remain difficult to avoid. This letter proposes a communication–localization fingerprint (CLF)-aided multi-UAV cooperative trajectory planning framework. An offline CLF map is first constructed to provide a unified spatial representation of communication and localization quality. The planning problem is then modeled as a decentralized partially observable Markov decision process (Dec-POMDP) and solved under a centralized-training decentralized-execution (CTDE) multi-agent reinforcement learning framework. Simulation results show that the proposed method steers UAVs away from service-degradation regions, improving communication–localization continuity and trajectory reliability.

Index Terms—Multi-UAV, communication–localization fingerprint, urban low-altitude environment, trajectory planning, integrated localization and communication, multi-agent reinforcement learning.

I. Introduction

Integrated localization and communication (ILAC) is a defining feature of sixth-generation (6G) wireless systems, reducing hardware cost [1] and improving spectral efficiency. Benefiting from line-of-sight (LoS) links, high mobility, and rapid deployment [2]–[6], uncrewed aerial vehicles (UAVs) have become key mobile platforms for ILAC in urban low-altitude environments. There, dense buildings block the air–ground links, posing serious challenges to the communication and localization performance of UAVs. To deliver reliable service in such environments, extensive research has investigated UAV trajectory planning and deployment for ILAC systems.

Optimization and heuristic approaches address UAV relay placement and 3-D position optimization [7] and joint deployment and communication–localization task allocation [8]. Affected by dynamic environments, however, these conventional algorithms incur high computational complexity and poor feasibility [9]. By contrast, reinforcement learning (RL) can improve its policy through environment interaction to reduce the complexity of the decision-making process [10], and has thus been increasingly adopted, spanning hierarchical multi-agent RL for joint resource and trajectory optimization [9], occlusion-aware trajectory and resource optimization [11], and joint trajectory and beamforming design [12]. Despite their effectiveness, these methods still fall short of ensuring communication and localization reliability in urban low-altitude environments: facing the abrupt service dead zones induced by building blockage, they often fail to detour in time, so UAVs slip into communication outage and localization lock-loss.

Jointly guaranteeing the two services is difficult because they impose different spatial requirements: communication needs only a single LoS base station (BS), whereas TDOA localization requires at least four well-distributed BSs and is highly sensitive to their geometry through the Cramér–Rao lower bound (CRLB). Their degradation regions therefore do not coincide, and the coupling is further compounded by multi-UAV airspace contention, energy limits, and collision avoidance. To sustain both, a planner must know in advance how each is distributed in space. This is echoed by radio- and channel-map-aided planning [13]–[15], where a pre-built spatial representation enables foresighted trajectory decisions.

Building on this idea, this letter proposes a communication–localization fingerprint (CLF)-aided cooperative trajectory planning method for urban multi-UAV ILAC systems. First, an offline CLF map is constructed, using Shannon capacity and the TDOA-CRLB to characterize communication and localization quality within a single normalized spatial representation, which provides the planner with look-ahead awareness of both services and their degradation regions. Second, the multi-UAV cooperative planning problem, subject to joint communication, localization, energy, and collision-avoidance constraints, is formulated as a decentralized partially observable Markov decision process (Dec-POMDP) and solved with a centralized-training decentralized-execution (CTDE) multi-agent RL strategy that enables cooperative planning with decentralized execution. Simulations in an urban blockage scenario show that the proposed method outperforms CLF-blind baselines on both communication and localization metrics, steering UAVs around service-degradation regions, and further reveal a tunable communication–localization tradeoff.

II. System Model and Problem Formulation

A. Urban Environment Model

Consider a typical urban low-altitude environment containing buildings and ground BSs. The building set

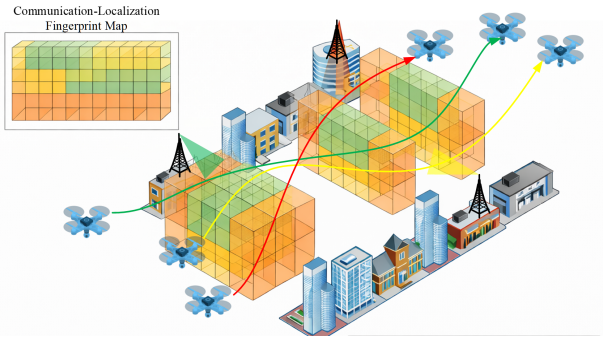


Fig. 1: Urban CLF-aided multi-UAV trajectory planning framework.

is denoted by $\mathbf{B} = \{b_1, b_2, \ldots, b_{|\mathbf{B}|}\}$, where building b_j is modeled as an axis-aligned cuboid with center $\mathbf{c}_j^b = (x_j^b, y_j^b, z_j^b)$ and dimensions (l_j, w_j, h_j). The BS set is denoted by $\mathbf{M} = \{m_1, m_2, \ldots, m_{|\mathbf{M}|}\}$, where BS m_k has position $\mathbf{p}_k^m = (x_k^m, y_k^m, z_k^m)$, transmit power P_k, and carrier frequency f_c.

To facilitate trajectory planning and environment modeling, the 3-D airspace is discretized into a voxel set $\mathbf{V} = \{v_1, v_2, \ldots, v_{|\mathbf{V}|}\}$. Voxel v_i has center $\mathbf{c}_i^v = (x_i^v, y_i^v, z_i^v)$, edge length Δr, and occupied region $\Omega(v_i)$. The occupied region of building b_j is denoted by $\Omega(b_j)$. Whether a voxel is occupied by a building is captured by the indicator function

$$I_{\text{occ}}(v_i) = \begin{cases} 1, & \exists b_j \in \mathbf{B},\ \Omega(v_i) \cap \Omega(b_j) \neq \varnothing, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

Considering obstacle avoidance, safety distance, and low-altitude flight constraints, the feasible voxel set is defined as

$$\begin{aligned} \mathbf{V}_{\text{feas}} = \{ & v_i \in \mathbf{V} \mid I_{\text{occ}}(v_i) = 0, \\ & \min_{b_j \in \mathbf{B}} d(\Omega(v_i), \Omega(b_j)) \geq d_{\min}^{\text{safety}}, \\ & h_{\min} \leq z_i^v \leq h_{\max} \}, \end{aligned} \tag{2}$$

where $d(\Omega(v_i), \Omega(b_j))$ is the minimum distance between voxel v_i and building b_j, $d_{\min}^{\text{safety}}$ is the safety-distance threshold, and $h_{\min}$ and $h_{\max}$ are the minimum and maximum UAV flight altitudes, respectively.

The multi-UAV set is denoted by $\mathbf{U} = \{u_1, u_2, \ldots, u_{|\mathbf{U}|}\}$. The start and target voxels of UAV u_n are v_n^{start} and v_n^{end}, respectively. At discrete time slot $t \in \{0, 1, \ldots, T\}$, the position of UAV u_n is represented by its hosting voxel $v_n(t) \in \mathbf{V}_{\text{feas}}$, and its trajectory is

$$\text{Traj}_n = \{v_n(0), v_n(1), \ldots, v_n(T)\}, \tag{3}$$

where $v_n(0) = v_n^{\text{start}}$, and the UAV is expected to reach v_n^{end} within the finite planning horizon. Fig. 1 illustrates the overall CLF-aided planning framework.

B. Communication–Localization Fingerprint

To jointly characterize spatial communication and localization performance in urban environments, we define the communication–localization fingerprint (CLF) of each feasible voxel $v_i \in \mathbf{V}_{\text{feas}}$ as

$$F(v_i) = \{f_{\text{comm}}(v_i), f_{\text{loc}}(v_i)\}, \tag{4}$$

where $f_{\text{comm}}(v_i)$ and $f_{\text{loc}}(v_i)$ are the communication and localization fingerprints, respectively.

A line-of-sight (LoS) link exists between v_i and BS m_k if the connecting segment is occlusion-free and within the maximum communication range. Let $\mathbf{M}_i$ denote the set of LoS-accessible BSs for voxel v_i.

1. Communication performance. For the link between v_i and m_k, the signal-to-noise ratio is

$$\text{SNR}_{i,k} = \frac{P_k G_{i,k}}{\sigma_0^2}, \tag{5}$$

where σ_0^2 is the noise power, $G_{i,k} = \beta_0\, d_{i,k}^{-\alpha}$ is the channel gain with $d_{i,k} = d(\mathbf{c}_i^v, \mathbf{p}_k^m)$, β_0 the reference gain, and α the path-loss exponent. The achievable capacity is $R_{i,k} = B \log_2(1+\text{SNR}_{i,k})$, where B is the system bandwidth. The communication performance of v_i is then

$$C(v_i) = \max_{m_k \in \mathbf{M}_i} R_{i,k}. \tag{6}$$

2. TDOA localization performance. For positioning reference signal (PRS)-based TDOA ranging, the TOA estimation error variance is $\sigma_{i,k}^2 = c^2 T_{\text{sym}}^2 / (8\pi^2\, \text{SNR}_{i,k} \sum_{j=1}^{N_J} p_j^2 j^2)$, where c is the speed of light, T_{sym} is the symbol duration, and p_j is the power weight of the j-th subcarrier. Taking BS m_1 as the reference anchor, the CRLB for 3-D TDOA localization is [16]

$$\text{CRLB}(v_i) = \text{tr}\big((\mathbf{D}_i^{\text{T}} \mathbf{R}_i^{-1} \mathbf{D}_i)^{-1}\big), \tag{7}$$

where $\mathbf{D}_i$ is the Jacobian matrix whose $(k-1)$-th row is $[(\mathbf{c}_i^v - \mathbf{p}_1^m)/d_{i,1} - (\mathbf{c}_i^v - \mathbf{p}_k^m)/d_{i,k}]^{\text{T}}$ for $k = 2, \ldots, |\mathbf{M}_i|$, and $\mathbf{R}_i$ is the covariance matrix with $[\mathbf{R}_i]_{k,k'} = \sigma_{i,1}^2 + \sigma_{i,k+1}^2 \delta_{k,k'}$ where $\delta_{k,k'}$ is the Kronecker delta. The localization error metric is $L(v_i) = \sqrt{\text{CRLB}(v_i)}$.

3. Normalized fingerprints. To unify the two metrics, we construct normalized fingerprints via sigmoid mapping. Let $S(x) = 1/(1+e^{-x})$ and define

$$\Phi(x; \tau, \alpha) = \frac{S(\alpha(x-\tau)) - S(-\alpha\tau)}{1 - S(-\alpha\tau)}. \tag{8}$$

The communication and localization fingerprints are then

$$f_{\text{comm}}(v_i) = \begin{cases} \Phi(C(v_i); C_{\text{th}} - \delta_C, \alpha_C), & |\mathbf{M}_i| \geq 1, \\ 0, & \text{otherwise,} \end{cases} \tag{9}$$

$$f_{\text{loc}}(v_i) = \begin{cases} \Phi(1/L(v_i); 1/(L_{\text{th}} + \delta_L), \alpha_L), & |\mathbf{M}_i| \geq 4, \\ 0, & \text{otherwise,} \end{cases} \tag{10}$$

where C_{th}, L_{th} are the communication and localization thresholds, and $\delta_C, \alpha_C, \delta_L, \alpha_L$ are the corresponding offset and slope parameters. When $|\mathbf{M}_i| < 4$, 3-D TDOA localization is infeasible, so $f_{\text{loc}}(v_i) = 0$.

C. UAV Mobility Model

In the discretized airspace, each UAV can only select its next voxel from the adjacent feasible voxel set $\mathbf{V}_n^{\mathrm{adj}}(t)$ of its current voxel $v_n(t)$ at each time slot. Consequently, a UAV trajectory is a sequence of discrete voxel centers. To respect flight speed constraints, the action set is restricted to the 26-neighborhood and hovering. For a cubic voxel with edge length Δr, the single-step displacement takes values in $\{0, \Delta r, \sqrt{2}\Delta r, \sqrt{3}\Delta r\}$, corresponding to face-adjacent, edge-adjacent, corner-adjacent moves, and hovering, respectively. The equivalent flight speed $V_n(t)$ is thus quantized to the finite set $\{0, \Delta r/\Delta t, \sqrt{2}\Delta r/\Delta t, \sqrt{3}\Delta r/\Delta t\}$.

For energy consumption, the propulsion power $P(V_n(t))$ follows the rotary-wing UAV power model in [17]. The single-step flight energy is $E_n^{\mathrm{fly}}(t) = P(V_n(t)) \cdot \Delta t$. Since $V_n(t)$ only takes a finite set of discrete values, the single-step energy of each action can be precomputed offline.

D. Multi-UAV Planning Problem

The multi-UAV cooperative trajectory planning problem aims to select, for each UAV at each time slot, a feasible next voxel such that the multi-UAV system achieves favorable communication performance, localization performance, and energy efficiency during trajectory execution while satisfying all constraints. The optimization problem is formulated as

$$\begin{aligned}
\max_{\{v_n(t)\}} \quad & \sum_{n=1}^{|\mathbf{U}|}\sum_{t=1}^{T}\Big[w_C f_{\mathrm{comm}}(v_n(t)) + w_L f_{\mathrm{loc}}(v_n(t)) - w_E E_n^{\mathrm{fly}}(t)\Big] \\
\text{s.t.} \quad & v_n(t) \in \mathbf{V}_{\mathrm{feas}} \cap \mathbf{V}_n^{\mathrm{adj}}(t-1), \quad \forall n, t, \\
& v_n(t) \neq v_{n'}(t), \quad \forall n \neq n', t, \\
& \{v_n(t-1), v_n(t)\} \neq \{v_{n'}(t), v_{n'}(t-1)\}, \ \forall n \neq n', t, \\
& v_n(0) = v_n^{\mathrm{start}}, \quad \exists\, t_n \leq T: \ v_n(t_n) = v_n^{\mathrm{end}}, \ \forall n, \\
& v_n(t) = v_n^{\mathrm{end}}, \quad \forall t \geq t_n.
\end{aligned} \tag{11}$$

Here, w_E, w_C, and w_L are the energy penalty, communication demand, and localization demand factors, respectively, with $w_C + w_L = 1$, and $E_n^{\mathrm{fly}}(t)$ is the single-step flight energy. The constraints respectively confine each UAV to a feasible voxel adjacent to its previous one, forbid two UAVs from sharing a voxel (co-voxel collision) and from swapping voxels across a slot (edge-swap collision), and require each UAV to reach its target at some $t_n \leq T$ and hold it thereafter.

Although the CLF map is constructed offline, each UAV can only access local information during the execution phase. The problem is therefore a partially observable multi-agent sequential decision-making problem. In the next section, we further model it as a Dec-POMDP and solve it using a CTDE-based MARL framework.

III. Proposed Algorithm

Under the CTDE framework, MARL algorithms are employed to solve the optimization problem formulated in Section II. This letter adopts multi-agent proximal policy optimization (MAPPO) as the concrete instantiation of the Dec-POMDP. The overall CTDE-based MARL architecture is illustrated in Fig. 2.

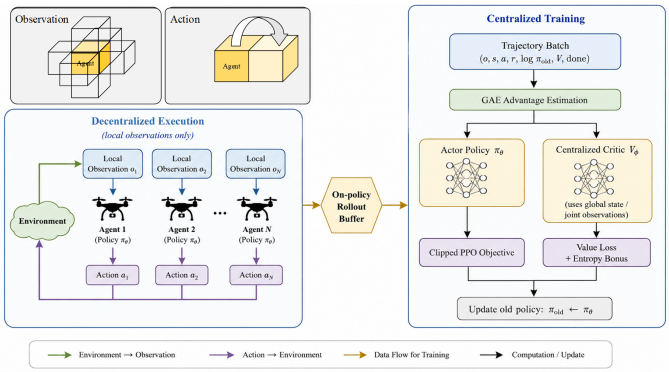


Fig. 2: CTDE-based MARL framework for CLF-aided multi-UAV trajectory planning.

The design is detailed as follows.

1. State space. At time slot t, the global state of the system consists of all UAVs' voxel positions and their corresponding CLF values, defined as

$$s(t) = \{v_n(t), F(v_n(t))\}_{n=1,\cdots,|\mathbf{U}|}. \tag{12}$$

During decentralized execution, the local observation of UAV u_n is defined as

$$o_n(t) = \{v_n(t), F(v_n(t)), \mathbf{V}_n^{\mathrm{adj}}(t), F(\mathbf{V}_n^{\mathrm{adj}}(t))\}, \tag{13}$$

where $\mathbf{V}_n^{\mathrm{adj}}(t)$ and $F(\mathbf{V}_n^{\mathrm{adj}}(t))$ are respectively the set of adjacent voxels of $v_n(t)$ and their fingerprint information.

2. Action space. Each UAV moves in the discrete grid space, and its action is defined as

$$a_n(t) \in \mathcal{A}_n = \{\mathbf{V}_n^{\mathrm{adj}}(t)\}. \tag{14}$$

The action essentially selects the next voxel position from the feasible adjacent voxel set of the current voxel, i.e., $v_n(t) \xrightarrow{a_n(t)} v_n(t+1)$.

3. Reward design. This letter constructs a team-averaged immediate reward function based on voxel fingerprint information. To ensure that all components are comparable under a common scale, the communication and localization fingerprints are directly used in their normalized forms, and the energy term is individually normalized. The immediate reward function is defined as

$$\begin{aligned}
r(t) = & \frac{1}{|\mathbf{U}|}\sum_{n}\Big\{X_n(t)\Big[w_C(f_{\mathrm{comm}}(v_n(t)) - 1) \\
& + w_L(f_{\mathrm{loc}}(v_n(t)) - 1) - w_E\widetilde{E}_n(t) - \delta_n^{\mathrm{coll}}(t)\Big] \\
& + r_{\mathrm{arr}}\big[X_n(t-1) - X_n(t)\big]\Big\}.
\end{aligned} \tag{15}$$

The first two lines form the per-step service–energy–safety cost, gated by the arrival indicator $X_n(t)$ so that a UAV stops accruing cost once it reaches its target. The last term is a one-off arrival bonus $r_{\mathrm{arr}} > 0$ granted at the single time slot when X_n transitions from 1 to 0 (i.e., the step on which UAV u_n first reaches v_n^{end}), which makes

Algorithm 1 CLF-Aided Multi-UAV Trajectory Planning via MAPPO

Input: CLF map $F(v_i)$, feasible voxel set $\mathbf{V}_{\text{feas}}$, start/goal voxels v_n^{start}, v_n^{end}, and MAPPO hyperparameters.
Output: Shared policy network π_θ.

```
Initialize shared actor θ and centralized critic φ.
repeat
    Reset UAV positions v_n(0) ← v_n^start and clear buffer D.
    for t = 0, ..., T − 1 do
        Each UAV constructs o_n(t) from local CLF observations and
        selects a feasible action a_n(t) ~ π_θ(· | o_n(t)).
        Execute joint action, obtain team reward r(t), observe s(t + 1),
        and store transition in D.
        if all UAVs reach their target voxels then
            break
        end if
    end for
    Compute GAE advantages Â(t) and returns R̂(t).
    Update actor θ using PPO clipped objective and update critic φ
    using value loss.
until episode = Epi_max
```

goal completion the optimal terminal behavior under the otherwise non-positive per-step reward. Here, $X_n(t)$ is an arrival-status indicator variable defined as

$$X_n(t) = \begin{cases} 1, & v_n(t) \neq v_n^{\text{end}}; \\ 0, & \text{otherwise}, \end{cases} \tag{16}$$

which indicates whether UAV u_n remains in an uncompleted mission state. When $X_n(t) = 0$, the UAV is considered to have reached its target voxel; its action is locked to hovering and it no longer accumulates communication fingerprint, localization fingerprint, or flight energy cost.

The normalized energy penalty is defined as $\bar{E}_n(t) = E_n(t)/E_{\max}$, where $E_n(t)$ is the single-step flight energy and $E_{\max}$ is the maximum single-step energy, corresponding to corner-adjacent movement.

$\delta_n^{\text{coll}}(t)$ is the penalty for multiple UAVs occupying the same voxel, defined as

$$\delta_n^{\text{coll}}(t) = \lambda_{\text{coll}} I_{\text{coll}}(v_n(t)), \tag{17}$$

where λ_{coll} is the collision penalty intensity and $I_{\text{coll}}(v_n(t))$ is the collision indicator function defined as

$$I_{\text{coll}}(v_n(t)) = \begin{cases} 1, & \exists n' \neq n, \text{ s.t. } v_n(t) = v_{n'}(t) \text{ or} \\ & v_n(t) = v_{n'}(t-1) \text{ and } v_n(t-1) = v_{n'}(t), \\ 0, & \text{otherwise.} \end{cases} \tag{18}$$

The collision indicator captures two types of conflicts: co-voxel occupancy and edge-swap crossing between adjacent time slots.

In summary, the reward function exploits the normalized nature of voxel fingerprints to achieve a unified metric for communication and localization performance, and constrains flight energy and flight safety under a common scale through the normalized energy term and the per-UAV collision term.

Based on the above state space, action space, and reward function designs, the MAPPO algorithm proceeds as follows.

IV. Experimental Evaluation

Simulations use the urban low-altitude scenario of Section II, with buildings, ground BSs, and UAVs restricted to feasible voxels. The communication and localization thresholds C_{th} and L_{th} are calibrated automatically from the medians of $C(v_i)$ and $1/L(v_i)$ over all feasible voxels, so that the sigmoid transition zones each span roughly 30%–70% of the feasible space. The navigation horizon and shaping coefficients are scaled with $1/\Delta r$ to maintain a consistent physical travel budget across resolutions. Key simulation parameters are summarized in Table I.

TABLE I: Key Simulation Parameters

Parameter	Value
Environment size	$600 \times 600 \times 150\,\text{m}^3$
Number of buildings	15
Number of BSs	12
Number of UAVs	3
Altitude range	$h_{\min} = 60\,\text{m}$, $h_{\max} = 120\,\text{m}$
Safety distance	$d_{\min}^{\text{safety}} = 5\,\text{m}$
Voxel size	$\Delta r = 20\,\text{m}$
System bandwidth	$B = 10\,\text{MHz}$
Carrier frequency	$f_c = 2.4\,\text{GHz}$
BS transmit power	$P_k = 30\,\text{dBm}$
Path-loss exponent	$\alpha = 2.4$
Noise power	$\sigma_0^2 = -80\,\text{dBm}$
Service weights	$w_C = 0.5$, $w_L = 0.5$
Discount factor	$\gamma = 0.99$
GAE parameter	$\lambda = 0.95$
PPO clip range	$\epsilon = 0.2$
Actor learning rate	3×10^{-3}
Critic learning rate	5×10^{-3}
Hidden layer width	128
Training episodes	4000

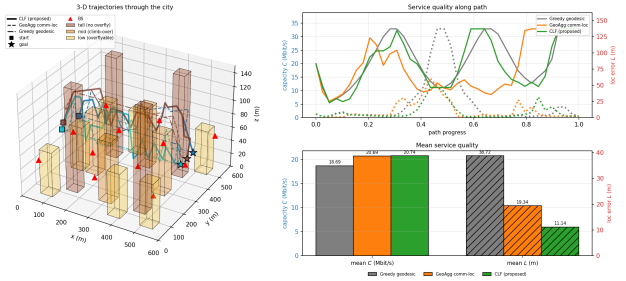
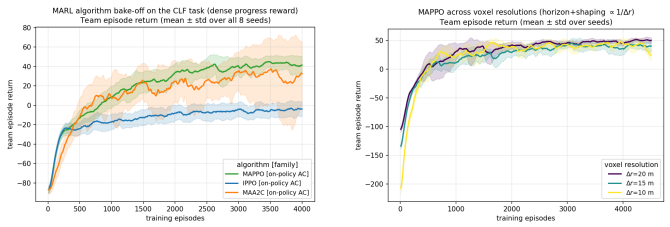


Fig. 3: Three-way method comparison. Left: top-down trajectories over f_{loc} field. Top-right: capacity C (solid) and localization error L (dotted) along path. Bottom-right: mean C and L comparison. Line styles distinguish methods; colors identify UAVs.

Fig. 3 compares the proposed CLF method against two baselines on neutral physical metrics—Shannon capacity C (Mbit/s) and localization error $L = \sqrt{\text{CRLB}}$ (m). The geodesic greedy baseline is CLF-blind, following geometry-only shortest paths. The GeoAgg baseline, adapted from [11], uses a log-potential reward that directly optimizes raw physical quantities. All three achieve full arrival with zero collisions. The proposed CLF method outperforms both baselines on both metrics. The geodesic baseline drives straight through the dead-zone, suffering severe localization degradation along the path. Lacking such look-ahead, GeoAgg neither anticipates nor avoids these degradation regions. In contrast, the CLF's thresholded, saturating reward balances both dimensions while its voxel-fingerprint look-ahead steers UAVs around the dead-zone.



(a) (b)

Fig. 4: Training dynamics: (a) algorithmic comparison (MAPPO, MAA2C, IPPO); (b) voxel resolution sweep ($\Delta r = 20/15/10\,\mathrm{m}$). Shading denotes ± 1 std. over seeds.

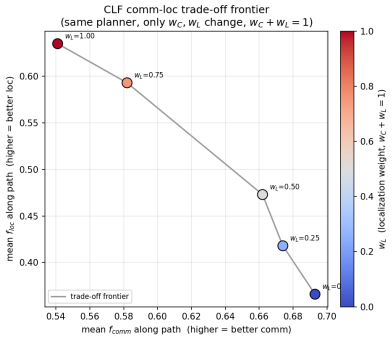


Fig. 5: Communication–localization tradeoff frontier obtained by sweeping w_L under $w_C + w_L = 1$. Marker color encodes w_L; the line traces the monotone frontier.

Fig. 4(a) isolates algorithmic factors. MAPPO achieves the highest team return with smallest inter-seed variance. IPPO, which drops the centralized critic, performs substantially worse, confirming centralized training is essential under partial observability. MAA2C reaches competitive average return but with larger variance, suggesting PPO clipping mainly stabilizes training. Fig. 4(b) examines resolution effects: the coarse grid ($\Delta r = 20\,\mathrm{m}$) converges fastest as its compact state–action space eases optimization; finer grids climb more slowly. Voxel resolution thus trades spatial fidelity against sample complexity: coarse grids suit limited training budgets or smoothly varying fields, finer grids higher-precision needs.

Fig. 5 shows how the CLF framework lets an operator navigate the communication–localization tradeoff: sweeping the single weight $w_L \in \{0, 0.25, 0.5, 0.75, 1.0\}$ under $w_C + w_L = 1$ (a fresh MAPPO trained per point) dials the service balance to demand. The resulting frontier is monotone—raising w_L from 0 to 1 lifts f_{loc} from 0.37 to 0.64 while f_{comm} falls from 0.69 to 0.54—and all UAVs reach their targets at every operating point, showing that this tunability comes at no cost to task completion.

V. Conclusion

This letter proposed a CLF-aided multi-UAV cooperative trajectory planning method for urban low-altitude environments. By combining an offline CLF map with CTDE-based MARL, the proposed method achieves cooperative planning under communication, localization, energy, and collision constraints. Simulation results show that, compared to CLF-blind baselines, the proposed CLF-aware MAPPO improves both the mean communication and localization fingerprints while all UAVs reliably reach their targets. The Pareto frontier analysis further reveals a tunable communication–localization tradeoff that lets an operator steer the service balance toward demand without sacrificing task completion. These results confirm that the proposed method effectively steers UAVs away from service-degradation regions, improving trajectory reliability and service continuity in urban blockage environments.

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